AN ABELIAN VARIETY IS SMOOTH OVER ITS BASE FIELD

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: this chapter's pool has 47 of 128 labels already delivered by the S7 line (S7-gp-* registrations, statement_anchor-prose scan; withdrawn cards excluded); none of the 11 accepted labels of this batch is among them (gp-occupied.json, 2026-10-14).

Source reference. Original lemma 047N: sources/stacks-chow/tags.txt, source lines 5514–5514. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 047P: sources/stacks-chow/tags.txt, source lines 5516–5516. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02VL: sources/stacks-chow/tags.txt, source lines 3772–3772. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BFB: sources/stacks-chow/tags.txt, source lines 14379–14379. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let A be an abelian variety over k. Then A is smooth over k.*

Proof. If k is perfect then this follows from Lemma 047N (characteristic zero) and Lemma 047P (positive characteristic). We can reduce the general case to this case by descent for smoothness (Descent, Lemma 02VL) and going to the perfect closure using Lemma 0BFB. □

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